

VizChitra

Physical Installation Brief

Direction A: Material / Tactile

"Data, Otherwise" Exhibition, 2026

Companion to the 65-inch Interactive Digital Screen Piece

70 Years of Ganga Basin Hydrological Data, 1951-2021

1,526 River Segments | Five Drainage Tiers | 362 Named Rivers

Working Title

"Still Flowing, Still Falling"

Generated: June 2026

1. Concept

A topographic relief map of the Ganga Basin carved in layered wood, where the physical depth of each river channel encodes its 70-year streamflow trend. Stable rivers sit at the surface. Drying rivers are cut progressively deeper into the wood. Teal resin pooled into the deepest channels marks statistically significant loss.

The visitor encounters a landscape where crisis is literally concentrated in specific corridors they can see and touch, while the majority of the surface remains whole and uncut -- making the spatial truth of "two-thirds still flowing" immediately legible through material contrast.

The piece functions as evidence: warm wood for the stable majority, teal depth for the drying minority, murky resin where record lows are recent. No animation, no electricity, no interaction required. The data is in the material itself.

2. Form Factor

Dimensions

120 cm (W) x 90 cm (H) x 8 cm (D), wall-mounted at 95 cm center height, angled 12 degrees from vertical for viewing comfort. The scale is large enough to resolve Tier 3-4 tributaries while remaining a single-person viewing object.

Primary Material

Stacked and laminated birch plywood (7-10 layers, each 6-8 mm), finished with a clear matte sealant to preserve warm wood grain. The layered cross-section is intentionally visible at the edges -- the strata echo the temporal layering of the data.

River Channels

CNC-milled into the layered surface. Five stroke widths corresponding to the five drainage tiers (Tier 1 mainstem widest at 6 mm, Tier 5 headwater narrowest at 1 mm). Channel depth varies from 0 mm (stable, no change) to 6 mm (severe drying, >50% flow loss).

Resin Inlay

Two-part epoxy resin tinted in a teal-to-amber gradient, poured into milled channels at varying depths. Drying segments receive teal resin fill; stable segments remain bare wood; the 16 wetting segments receive a thin amber resin line at the surface.

Frame

Blackened steel L-bracket frame, floating 3 cm off the wall with concealed mounting hardware. The shadow gap adds visual depth and separates the piece from the gallery wall.

Label Panel

Separate adjacent panel (40 cm x 60 cm) in the same birch, laser-engraved with legend, data citation, narrative text, and a QR code linking to the interactive digital piece.

3. Data Encoding

Every physical property of the piece maps to a specific data variable. The encoding is designed so that the spatial pattern is legible before any legend is read -- the visitor sees and feels the concentration of depth before understanding what it means.

Physical Property	Data Variable	Encoding Rule
Channel depth (0-6 mm)	A1 percent change in streamflow, 1951-2021	0% change = surface (0 mm). Each 10% decline = ~1 mm deeper. Segments with >50% decline reach max 6 mm depth. Wetting segments also sit at surface -- loss is encoded as depth.
Channel width (1-6 mm)	Tier (1-5)	Tier 1 mainstem = 6 mm wide. Tier 5 headwater = 1 mm wide. Fixed geographic anchor, identical to the digital piece's stroke width.
Resin color	A6 classification (drying / stable / wetting)	Drying: teal resin filling the channel. Stable: no resin -- bare wood visible. Wetting (16 segments): thin amber resin line at surface. Dominant contrast is wood (stable) vs. teal-filled (drying).
Resin opacity / pooling	B2 minimum year (pre- vs. post-1991)	Post-1991 minima receive a second, slightly clouded resin layer -- channels appear deeper and less transparent. Encodes the floor-drop: resin is murkier where record lows are recent.
Cluster visibility	Spatial concentration of drying	Geographic encoding makes clusters self-evidencing. Yamuna corridor (86.8% drying), Sai-Gomti belt (100% drying), upper Ganga emerge as teal valleys. Chambal, Son, eastern tributaries remain warm bare wood.

Design Integrity Note

The encoding preserves the project's core narrative structure. The stable majority (65.1%) dominates the visual field as warm, uncut wood -- preventing the piece from reading as catastrophe. The drying corridors are visible but concentrated, matching the data's actual geography. The B2 floor-drop layer (clouded resin) adds the second tension without overriding the first: most channels are shallow, but those that are deep are also murky.

4. Visitor Experience

Approach (3+ meters): Aesthetic Register

The visitor sees a beautiful topographic wood carving -- a river network rendered in warm birch with teal accents. The form reads as decorative or cartographic. The material warmth invites approach.

Mid-distance (1 meter): Pattern Recognition

The uneven depths become visible. Some channels are shallow surface grooves; others are cut deep enough to pool with teal resin. The Yamuna corridor and Sai-Gomti belt stand out as concentrated teal valleys. The eastern tributaries (Son, Kosi) remain barely disturbed. The spatial concentration of crisis registers before any label is read.

Close Range (arm's reach): Tactile Exploration

The visitor traces a finger along the Yamuna corridor -- the drop is palpable. Moving east to the Son, the

surface is barely incised. The hand learns what the data says: the crisis is real but concentrated. The murky resin in the deepest channels is visible up close, encoding the recency of record lows.

Reading (label panel): Data Comprehension

The legend provides the key: depth = flow loss, teal resin = statistically significant drying, bare wood = stable, clouded resin = post-1991 record low. The visitor re-reads the piece with this knowledge. A QR code links to the interactive digital piece for temporal exploration.

Expected Dwell Time

2-4 minutes. Aesthetic approach (~30 sec), pattern recognition (~30 sec), tactile exploration (~1 min), legend reading and re-examination (~1-2 min). Visitors who then move to the digital screen carry the spatial memory of where the deep channels were.

5. Relationship to Digital Piece

Complementary, not standalone.

The physical piece and the 65-inch interactive screen serve different registers of the same data story. Neither duplicates the other.

Dimension	Digital Screen	Physical Piece
Primary sense	Sight (animated color)	Touch + sight (depth, texture)
Temporal vs. spatial	Unfolds 70 years over time (three-beat animation arc)	Collapses 70 years into a single static object -- the where is immediately graspable
Narrative beat emphasized	All three beats: stability, breathing rhythm, floor-drop	Beat 1 (stability as dominant surface) + spatial concentration of drying corridors
Interaction required	Must stop, watch, scrub, pause, hover for full experience	Always legible -- works for a glance from across the room and for five minutes of tracing
Material register	Screen-based, narrative, temporal, rhythmic	Wood and resin feel archival, permanent, geological -- evidence, not animation
B2 floor-drop encoding	Temporal accumulation (marks appear over animation years -- surprise builds)	Material opacity (clouded resin in post-1991 channels -- visible only on close inspection)

A visitor who has touched the Yamuna valley in the wood piece will recognize it instantly on the digital screen when the animation reaches the 2000s and that corridor floods amber. The physical memory of depth becomes the context for the temporal story of change.

6. Fabrication Notes

CNC Milling

Requires a 3-axis CNC router with a fine-detail bit (1 mm end mill for Tier 5 channels). The river network geometry can be exported from the project's existing shapefiles (Shapefiles/Ganga Basin Streams/). A fabricator experienced with topographic map milling is ideal. Architectural model shops or topographic map fabricators are the right profile.

Resin Pour

Multi-stage pour: one clear teal layer for base depth, one slightly clouded layer in post-1991 minimum segments. Requires careful masking of stable segments (993 of 1,526) to keep them resin-free. A resin artist or experienced woodworker with epoxy experience is needed. This is the highest-risk fabrication step.

Data Preparation

Generate a CNC toolpath file mapping each of the 1,526 segments to its milled depth (from A1 percent change) and width (from tier). Export segment geometries with attributes attached, convert to SVG/DXF with depth metadata. Estimated 2-3 days of scripting.

Timeline

Phase	Duration	Description
Data preparation	1 week	Scripting: shapefile to CNC toolpath with depth/width attributes
CNC programming	1 week	Toolpath optimization, test cuts on scrap plywood (20-30 segments)
Milling	1 week	Full piece milling on laminated birch plywood panel
Resin pour & curing	1 week	Masking, two-stage pour (teal base + clouded B2 layer), 48hr cure
Finishing & mounting	1 week	Sanding, sealant, steel frame fabrication, label panel engraving

Total Lead Time: 4-6 weeks from final data file to finished piece.

Estimated Cost Range

\$2,500-\$4,500 (materials + fabrication). Breakdown: birch plywood and lamination (~\$300), CNC shop time (~\$800-1,200), resin materials (~\$200), steel frame (~\$400-600), fabrication labor (~\$800-1,500), label panel (~\$200-300).

Risk Mitigation

The main risk is the resin masking step -- keeping 993 stable channels resin-free while filling 517 drying channels. A test panel with 20-30 representative segments (mixed tiers, mixed depths, adjacent stable/drying channels) is strongly recommended before the full piece. Budget one extra week if the test panel reveals masking issues.

7. Data Grounding

Every design decision in this brief traces back to a specific data finding. The following table maps the key numbers that drive the physical encoding.

Finding	Source	Physical Expression
65.1% of segments show no significant long-term trend	A2, A6	The majority of the surface is uncut warm wood -- stability dominates the visual and tactile field
33.9% drying, concentrated in Yamuna, Sai-Gomti, upper Ganga	A6, A1	Teal resin pools in specific corridors, not uniformly across the basin

Finding	Source	Physical Expression
74.6% of segments hit all-time minimum flow after 1991	B2	Clouded second resin layer in post-1991 channels -- visible only on close inspection
Yamuna: 86.8% drying, -21.3% median change	A6, A1	Deepest, widest teal-filled valley on the piece
Sai/Sengur/Karwan: 100% drying, 0 dams	A6	Fully teal-filled channels with no interruption -- continuous depth
Chambal: 0% drying Son: 0% drying	A6	Bare wood surface -- no resin, no depth, warm and undisturbed
Dam-free segments dry faster (36.6%) than dammed (31.1%)	A6	Piece does not encode dams directly -- avoiding the false narrative that dams are the primary driver

The piece should feel like evidence, not elegy. The warmth of the wood holds the reassurance; the depth of the resin holds the concern. The visitor's hand discovers what the data already knows: most of this basin is still flowing, and the places that aren't are specific enough to name, to trace, to touch.

Data: mizuRoute hydrological model, 1951-2021. Analysis: Mann-Kendall trend tests,
Standardised Streamflow Anomaly, percent change, segment minima.
Scientific basis: Singh-Chuphal et al. (2025)